

Nima Jafari

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Open to full-time and contract-to-hire opportunities

Engineer building simulation-based robotics, machine learning, and full-stack AI systems. Excited to develop new systems with mission-oriented teams. Stanford SRC 2026 Best Paper Award in Computational Sciences and Mathematics.

Technical Skills

Machine Learning & AI: PyTorch, TensorFlow/Keras, reinforcement learning (PPO, Stable-Baselines3), supervised learning, clustering, computer vision, RAG, LLM application pipelines

Robotics & Simulation: NVIDIA Isaac Sim, USD scenes, Franka Panda, simulation environments, observation and controller integration, closed-loop evaluation, reproducible experiments

Software: Python, Java, C++, JavaScript/TypeScript, Swift, SQL, React, React Native, SwiftUI, Next.js, Node.js/Express, FastAPI

Infrastructure & Data: AWS, Azure, Docker, CI/CD, PostgreSQL, MongoDB, Neo4j, Redis, Firebase, NumPy, Pandas, scikit-learn

Experience

Machine Learning Engineer | Contract Projects

Jan 2023 – Sep 2026

Projects supported by DoW and NSF grants, including DARPA and AFOSR programs; appointment administered through University Corporation at CSUN.

- Built a Franka Panda simulation environment in NVIDIA Isaac Sim for tabletop manipulation, including planar pushing and guarded peg insertion; implemented structured observations, controller bindings, bounded closed-loop execution, and auditable trial logs.
- Developed and evaluated an Isaac study comparing wiring-diagram planning with a symbolic baseline: 29/30 original nominal trials reached target, and all 60/60 missing or stale observation fault trials stopped safely.
- Created reproducible benchmark tooling with held-out task families, ordered trial records, raw evidence, and configuration hashes; results are reported within the frozen study scope.
- Developed a from-scratch multinomial linear next-action policy baseline with separate training, validation, and evaluation families; qualified legal proposals and safe-stop behavior without direct robot-dispatch authority.
- Developed Hasse-diagram clustering for sequential reinforcement-learning behavior, recovering two winning strategies across all 125 successful episodes and evaluating robustness against 10% data corruption.
- Separately trained and evaluated CNN-based figure-skating video classifiers, including transfer learning with a Kinetics-pretrained R(2+1)D-18, and tested pose models including YOLOv8-Pose, MediaPipe, Keypoint R-CNN, and OpenPose.
- Co-authored two arXiv preprints during this appointment and presented findings at CSUN, Stanford SRC 2026, and the AFOSR 2025 Computational Cognition & Machine Intelligence Program Review.

Founding AI Engineer | Driftless

Aug 2025 – Present

Equity-based, part-time role at a pre-seed startup

- Helped build Driftless, a multi-tenant SaaS product for human teams and AI agents to coordinate work through shared task management and agent execution workflows.
- Delivered product features across the React client, Node.js/Express services, MongoDB data layer, and command-line agent tools, including task workflows, agent sessions, and organization-scoped access.
- Built an interactive terminal experience for agent runs with live task status, inbox handoffs, model/provider selection, onboarding, and resilient process lifecycle handling.
- Integrated agent and orchestrator workflows through scoped token bootstrap, model discovery, live session state, and guarded task execution paths.
- Improved responsive web and mobile workspaces while contributing to production CI, automated quality gates, and release workflows.

Full-Stack iOS Engineer (Contract) | SideShift NYC

May 2026 – May 2026

- Shipped SwiftUI interface fixes and backend-supported web features; improved data access and caching to reduce unnecessary cloud database and API usage.
- Collaborated on product refinements related to subscription conversion and premium feature adoption.

- Supported firefighting and emergency-response robotics through software development, hardware assembly, mechatronic integration, and prototype testing.

Selected Projects

PocketPilot — Personal Finance App

- Built a full-stack personal finance app with an Expo/React Native mobile client for iOS and Android and a React web app.
- Connected mobile and web clients to Firebase-backed services and Rust-based transaction categorization and retrieval-augmented generation services.

Agentic AI — Policy-Grounded Support Agent

- Built a local-first e-commerce support agent for refund requests with a FastAPI backend, Streamlit interface, and configurable local or hosted language-model providers.
- Kept refund outcomes under deterministic policy control; protected actions require backend-issued decision IDs, with traces for review and audit.

Hasse Clustering for Reinforcement-Learning Game Behavior | GitHub repository

- Designed two versions of a video game and trained PPO reinforcement-learning agents to play them; applied Hasse-DAG clustering to gameplay traces to recover the games' winning strategies, comparing results with DBSCAN and hierarchical clustering.

CareerLift — AI Career Platform

- Built a full-stack platform for resume parsing, job matching, and interview coaching, with a Neo4j-backed RAG pipeline and local LLM inference. (Next.js, React, TypeScript, FastAPI, Python, Neo4j, LangChain, Ollama)

Reel Me Listing — AI Real Estate Video Studio

- Built a browser-based media pipeline that turns property photos into reviewed vertical listing reels, with image-quality checks, optional AI image edits, and video-generation workflows.
- Developed a React studio and local FastAPI service with reproducible job records, source tracking, and generated artifact review.

Education

California State University, Northridge

Fall 2022 – Spring 2026

B.S. in Computer Science | GPA: 3.98/4.00

Publications

- Jason Lo and Mohammadnima Jafari, "Wiring diagram extraction and gluing: a case study in classifying figure skating jumps using 3D dataset," arXiv:2607.27598, 2026.
- Jason Lo and Mohammadnima Jafari, "From data to concepts via wiring diagrams," arXiv:2511.20138, 2025.

Honors & Leadership

- Stanford SRC 2026 — Best Paper, Computational Sciences and Mathematics.
- CSUN CSE scholarships: Guerrero Endowed (2023), Engineering Merit (2024), Trustee Steven G. Stepanek Endowed (2025); IAWF scholarship awardee (2024, 2025).
- Co-founded the CSUN Alpha Lambda Delta chapter; served as Treasurer in 2024. Dean's List each semester from Fall 2022 through graduation in May 2026.